

Week 5 - Class 9: Control Flow Basics and User Input

Why This Matters

Until now, code has run top-to-bottom with no decisions or input. Real programs must:

- Ask users for information.
- Branch depending on input.
- Handle multiple options.

This class introduces user input (cin, getline) and control flow (if, else, switch).

Getting User Input with cin

Use cin (console input) to read values typed by the user.

Example:

```
int age;
cout << "Enter your age: ";
cin >> age;
cout << "You entered: " << age << "\n";
```

Multiple Inputs

You can read multiple values in one line:

```
int a, b;
cin >> a >> b; // enter: 5 10
cout << "a=" << a << ", b=" << b << "\n";
```

Strings and cin

cin >> works with strings but stops at whitespace.

```
string name;
cout << "Enter your first name: ";
cin >> name;
cout << "Hello, " << name << "!\n";
```

getline for Full Lines

Use getline() to read entire lines (with spaces).

```
string fullName;
cout << "Enter your full name: ";
getline(cin, fullName);
cout << "Hello, " << fullName << "!\n";
```

Mixing cin and getline

After cin >>, leftover newline may cause getline to read empty line.

Fix with cin.ignore().

```
int age;
cin >> age;
cin.ignore(numeric_limits<streamsize>::max(), '\n'); // clear newline
string fullName;
getline(cin, fullName);
```

Common Mistakes (Input)

1. Forgetting cin.ignore() when mixing >> and getline.
2. Assuming cin>> reads full lines (it stops at whitespace).
3. Not prompting user before input.
4. Forgetting to flush output when needed (use << endl or << flush).

if Statement

Syntax:

```
if(condition){ statements }
```

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Example:

```
int health = 0;
if (health == 0) {
    cout << "Game Over!\n";
}
```

else and else if

else runs when condition is false. Chain with else if for multiple branches.

```
if (grade >= 90) {
    cout << "A\n";
} else if (grade >= 80) {
    cout << "B\n";
} else {
    cout << "Fail\n";
}
```

Conditions and Booleans

C++ conditions evaluate to bool. 0 is false, nonzero is true. Best practice: use explicit bools.

switch Statement

switch compares one integral expression to constants. Use break; to avoid fall-through.

```
int choice = 2;
switch(choice) {
    case 1: cout<<"Start Game\n"; break;
    case 2: cout<<"Options\n"; break;
    default: cout<<"Invalid\n"; break;
}
```

Common Mistakes (Control Flow)

1. Forgetting break; causes fall-through.
2. Using non-integral in switch (not allowed).
3. Writing if(x=5) instead of if(x==5).
4. Assuming if(x) means positive; it means x!=0.

Exercises

1. Ask for two numbers and print their sum.
2. Ask for name (getline) and greet user.
3. Ask for age, branch into child/teen/adult.
4. Write switch mapping 1-7 to weekdays.
5. Try leaving out break; to see fall-through.

Project Tie-In

Expand Control Flow Demo:

- Ask user for number 1-10 (cin).
- Use if/else to say first half or second half.
- Use switch to print number as word.
- Use getline to read full name and personalize output.